Mohammad Arham

Odoo / Python Developer · ERP Customization & Integration · Cloud-Native Backend

mohdarham94@gmail.com • +91 9557385029 • Gurugram, Haryana • LinkedIn • GitHub • Portfolio

PROFESSIONAL SUMMARY

Odoo/Python Developer with 2 years of experience designing, developing, and customizing cloud-native ERP solutions using Odoo framework (v12/v16/v18). Proficient across the full SDLC — from techno-functional analysis to containerized production deployment. Deep expertise in Odoo ORM, QWeb, XML/JavaScript UI, PostgreSQL, REST API integration, and DevOps practices. Experienced with AI-assisted development (Codex, Cursor), LLM API integration, and OIDC/SSO & 2FA implementation. Delivered 60% database size reduction, 15-minute ETL latency, and 55% improvement in pipeline observability. Skilled at translating complex business requirements into scalable ERP solutions with deep third-party integrations.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Odoo: Odoo (v12/v16/v18), Odoo ORM, QWeb Templates, Scheduled Actions (ir.cron)

Frameworks: Flask, FastAPI, SQLAlchemy, Pydantic, DRF

Cloud & DevOps: AWS (EC2, S3, Lambda, SNS, Secrets Manager, Athena, Redshift), Docker, Jenkins, Nginx, CI/CD

Databases: PostgreSQL, MySQL, Amazon Redshift, Redis, SQLite

Data & Pipelines: Apache Airflow, ETL/ELT Pipelines, Incremental Sync, Data Ingestion, Task Scheduling, AWS Glue, Google BigQuery

API & Security: REST APIs, JWT, OAuth 2.0, RBAC, Rate Limiting, SSL/TLS, CORS, HTTPS, HMAC Authentication

Architecture: Microservices, Cloud-Native Design, OOP & Object Modeling, DB Query Optimization, System Design, Distributed Systems

Tools: Git, GitHub, JIRA, Postman, Swagger/OpenAPI, Codex, Linux

WORK EXPERIENCE

Binary Semantics Ltd | Associate Software Developer **Aug 2024 – Present**

Full-time · Gurugram, India · Odoo (v12/v16/v18), Python, Flask, FastAPI, PostgreSQL, Docker, AWS

- **Backend & API Development:** Developed and deployed full-stack cloud-native applications on the Odoo framework alongside Flask and FastAPI; applied OOP principles and object modeling with relational databases (PostgreSQL) to build secure, high-performance RESTful endpoints with JWT, RBAC, OAuth 2.0, and API rate limiting consumed by analytics and ERP platforms.
- **Data Engineering & ETL:** Architected a config-driven, incremental ETL pipeline using Python and Apache Airflow to replicate Odoo PostgreSQL data to Amazon Athena and Redshift; reduced data latency from daily batch loads to near real-time 15-minute intervals, enabling BI dashboard freshness for executive reporting.
- **Cloud Infrastructure & Optimization:** Migrated binary file storage from PostgreSQL to AWS S3, shrinking database size by ~60% and significantly improving query response times; deployed all services on EC2 behind Nginx with SSL via Certbot and custom domain routing.
- **ERP Customization (Odoo):** Designed, developed, and customized Odoo modules (v12/v16/v18) using Python ORM, QWeb templates, and XML views to automate and streamline core business processes; integrated Odoo with third-party systems via REST APIs, reducing manual intervention and improving ERP workflow reliability across finance, logistics, and CRM modules.
- **GPS Live Tracking Module (Odoo):** Designed and built a GPS live-tracking module in Odoo 12 with HMAC-authenticated JSON/HTTP API controllers, scheduled ir.cron jobs, and BigQuery integration to track driver trips in real time and share live location links via consignment notes; integrated end-to-end with a Flutter driver app and customer-facing portal, including trip-event push notifications and proof-of-delivery handling.
- **Third-Party & Real-Time Integrations:** Integrated ULIP, PAN, and DL verification APIs for KYC and finance compliance; implemented Google Maps API for real-time FASTag vehicle tracking in the logistics module, improving operational visibility.

- **DevOps & CI/CD:** Containerized all backend services with Docker; built Jenkins CI/CD pipelines for automated testing and zero-downtime deployments; managed secrets via AWS Secrets Manager; automated database dump workflows via Airflow with AWS SNS failure alerts.
- **AI & Chatbot Integration:** Integrated a Rasa NLU-powered chatbot with core backend APIs via REST webhooks, mapping user intents to business logic actions; actively leveraged AI-assisted development tools (GitHub Copilot, Cursor) to accelerate delivery and maintain code quality across sprints.

PROJECTS

Real-Time GPS Live Tracking System

Odoo 12 · Python · BigQuery · REST APIs

- Designed and built a GPS live-tracking module in Odoo 12 — custom JSON/HTTP API controllers with HMAC-signed authentication, scheduled ir.cron jobs, and BigQuery integration — to track driver trips in real time and share live location links with customers through consignment notes.
- Integrated the tracking system end-to-end with a Flutter driver app and customer-facing web portal, including automated push notifications for trip start/stop events and proof-of-delivery document handling tied to each consignment.

Inlog Analytics Dashboard — Enterprise BI Platform

Flask · Python · PostgreSQL (Raw SQL) · CanvasJS · JavaScript · Odoo ORM · AWS EC2 · Docker · RBAC

- Architected a Flask-based BI platform with 5 interactive dashboards (Sales, Marketing, HR, Logistics, Helpdesk) integrated with Odoo PostgreSQL; dynamic filtering engine for date, team, and employee segmentation with RBAC-enforced role-specific access.
- Authored complex SQL queries with CTEs and aggregations to power drilldown KPIs, trend reports, and performance benchmarks across all business units.

FastAPI Microservice — Real-Time Analytics API

FastAPI · Async Python · Pydantic v2 · PostgreSQL · AWS EC2 · Nginx · Docker

- Engineered a high-throughput async FastAPI microservice powering real-time analytics dashboards; leveraged Pydantic v2 schema validation, PostgreSQL connection pooling, and optimized indexed queries to achieve sub-50ms p95 response times.
- Deployed behind Nginx on AWS EC2 with HTTPS and health-check endpoints; reduced average database query time by ~40% through selective indexing and query plan analysis.

Config-Driven ETL Pipeline — Apache Airflow & AWS Athena

Python · Apache Airflow · AWS Athena · Redshift · PostgreSQL · AWS SNS

- Designed a modular, YAML-config-driven ETL framework with Airflow DAGs supporting incremental sync, full backfill, and automatic schema evolution for upstream Odoo data into AWS Athena and Redshift.
- Added SLA breach alerting via AWS SNS and Airflow callbacks; improved data-team incident response time by ~60% — pipeline now serves as the primary data source for 3 internal BI dashboards.

EDUCATION

Jamia Hamdard University — B.Tech in Computer Science **2020 – 2024 | CGPA: 8.19 / 10.0**

New Delhi, India · Relevant: Data Structures & Algorithms, Database Management, Operating Systems, Software Engineering, Computer Networks

CORE COMPETENCIES & KEYWORDS

Python • Odoo Developer • Odoo Framework (v12/v16/v18) • ERP Customization • QWeb • XML/HTML/CSS • JavaScript • Backend Development • REST API • FastAPI • Flask • Cloud-Native Applications • AWS • Docker • Kubernetes • CI/CD • DevSecOps • Data Engineering • ETL Pipeline • Apache Airflow • Google BigQuery • PostgreSQL • MySQL • Amazon Redshift • JWT Authentication • OAuth 2.0 • RBAC • OOP • Object Modeling • System Design • Database Optimization • SQLAlchemy • Pydantic • GitHub Copilot • Cursor • JIRA • Agile • Scrum • Third-Party Integration • Full Stack Development • Microservices • Real-Time Analytics • GPS Tracking • HMAC Authentication • Flutter • ir.cron